

Homework #8

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Question 7.

(a): 6.1.5

B.

$$\frac{\binom{13}{1}\binom{4}{3}\binom{12}{2}\binom{4}{1}\binom{4}{1}}{\binom{52}{5}}$$

C.

$$\frac{\binom{4}{1}\binom{13}{5}}{\binom{52}{5}}$$

D.

$$\frac{\binom{4}{2}\binom{13}{1}\binom{12}{3}4^3}{\binom{52}{5}}$$

(b): 6.2.4

A.

$$1 - \frac{\binom{39}{5}}{\binom{52}{5}}$$

B.

$$1 - \frac{\binom{13}{5}4^5}{\binom{52}{5}}$$

C.

$$\frac{2\binom{13}{1}\binom{39}{4} - \binom{13}{1}^2\binom{26}{3}}{\binom{52}{5}}$$

D.

$$1 - \frac{\binom{26}{5}}{\binom{52}{5}}$$

Question 8.

(a): 6.3.2

A.

$$\begin{aligned}P(A) &= \frac{1}{7} \\P(B) &= \frac{1}{2} \\P(C) &= \frac{5!}{7!} = \frac{1}{42}\end{aligned}$$

B.

$$P(A|C) = \frac{1}{10}$$

C.

$$P(B|C) = \frac{1}{2}$$

D.

$$P(A|B) = \frac{1}{7}$$

E.

(A, B) and (B, C) are independent while (A, C) is not.

(b): 6.3.6

B.

$$\left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^5$$

C.

$$\left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^9$$

(c): 6.4.2

A.

$$\approx 40.38\%.$$

Question 9.

(a): 6.5.2

A.

$$\{0, 1, 2, 3, 4\}$$

B.

$$\begin{aligned} &(0, C(48, 5)/C(52, 5)) \\ &(1, C(4, 1)C(48, 4)/C(52, 5)) \\ &(2, C(4, 2)C(48, 3)/C(52, 5)) \\ &(3, C(4, 3)C(48, 2)/C(52, 5)) \\ &(4, C(4, 4)C(48, 1)/C(52, 5)) \end{aligned}$$

(b): 6.6.1

A.

$$(0 * 3/45 + 21/45 + (2 * 21/45)) = 1.4$$

(c): 6.6.4

A.

$$E[X] = (1 + 4 + 9 + 16 + 25 + 36)/6 \approx 15.1666667$$

B.

$$E[Y] = 3/8 + 4(3/8) + 9(1/8) = 3$$

(d): 6.7.4

A.

$$10 * \frac{1}{10} = 1$$

Question 10.

(a): 6.8.1

A.

18.5%

B.

26.4%

C.

1

D.

39.5% and 1, the probabilities increase while the expected values stay the same.

(b): 6.8.3

B.

35.04%.